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Project Title :	**Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions**
Week :	Week 6

WEEK-6 – Multi-Channel Marketing Attribution

Task Name

Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions

Objective:

To analyze customer journeys across different marketing channels using a **Markov Chain** and identify which channels contribute most to conversions using the **Removal Effect**.

Problem Statement:

Customers interact with multiple marketing channels before making a purchase. It is difficult to determine which channel has the greatest impact on the final conversion.

This project uses **15 million user path logs** to:

1. Build a transition probability matrix.
2. Model customer movement between marketing channels.
3. Calculate the probability of conversion.
4. Remove each channel one by one.
5. Measure the reduction in conversion probability.
6. Rank channels according to their contribution.

Example Marketing Channels

Channel	Example
Organic Search	Google search
Paid Search	Search advertisements
Social Media	Instagram/Facebook

Channel	Example
Email	Promotional emails
Display Ads	Banner advertisements
Direct	Direct website visit
Conversion	Purchase
Null	No conversion

Markov Chain Model:

A customer journey may look like:

Start → Social Media → Organic Search → Email → Direct → Conversion

Each transition has a probability based on the observed user logs.

For example:

$$P(Social \rightarrow Email) = \frac{\text{Social → Email transitions}}{\text{Total Social transitions}}$$

All these probabilities form the **Transition Probability Matrix**.

Removal Effect

The Removal Effect measures how important a marketing channel is.

$$RemovalEffect_i = \frac{P(C) - P(C \mid i\,removed)}{P(C)}$$

Where:

- $P(C)$= conversion probability with all channels
- $P(C \mid i\,removed)$= conversion probability after removing channel i

A **higher Removal Effect** means that channel has a stronger contribution to conversions.

Example

Suppose:

- Original conversion probability = **0.20**
- Conversion probability after removing Email = **0.14**

$$RemovalEffect = \frac{0.20 - 0.14}{0.20} = 0.30$$

So the Email channel has a **30% removal effect**.

Expected Workflow:

15 Million User Logs
↓
Data Cleaning & Preprocessing
↓
Extract User Paths
↓
Count Channel Transitions
↓
Build Transition Probability Matrix
↓
Calculate Baseline Conversion Probability
↓
Remove Each Channel
↓
Recalculate Conversion Probability
↓
Calculate Removal Effect
↓
Rank Marketing Channels

Expected Output:

Marketing Channel	Conversion Probability After Removal	Removal Effect
Email	0.14	30%
Social Media	0.16	20%
Paid Search	0.17	15%
Organic Search	0.18	10%
Display Ads	0.19	5%

The values above are illustrative; actual values must be calculated from the 15-million-log dataset.

Conclusion:

This project uses **Markov Chain attribution** to understand the contribution of different marketing channels. The **Removal Effect** helps identify channels that have the greatest influence on customer conversions, allowing businesses to make better marketing-budget decisions.